claire_model

Load data

```
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.2     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.1
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(mgcv)
```

```
Loading required package: nlme
```

```
Attaching package: 'nlme'
```

```
The following object is masked from 'package:dplyr':

    collapse
```

```
This is mgcv 1.9-1. For overview type 'help("mgcv-package")'.
```

```
train <- read_csv("data-train.csv")
```

```
Rows: 89 Columns: 7
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl (7): St, Re, Fr, R_moment_1, R_moment_2, R_moment_3, R_moment_4

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
train <- train |>
  mutate(
    m1 = R_moment_1, m2 = R_moment_2,
    m3 = R_moment_3, m4 = R_moment_4,

    # Central moments
    C_moment_2 = m2 - m1^2,
    C_moment_3 = m3 - 3*m1*m2 + 2*m1^3,
    C_moment_4 = m4 - 4*m1*m3 + 6*m1^2*m2 - 3*m1^4,

    # Summary statistics
    sd_R = sqrt(C_moment_2),
    skewness = C_moment_3 / sd_R^3,
    kurtosis = C_moment_4 / sd_R^4,
    ex_kurt = kurtosis - 3,

    # Keep infinite Fr as a valid category
    Fr = as.factor(Fr),
    Re = as.factor(Re)
  )
```

Fit a spline model for mean

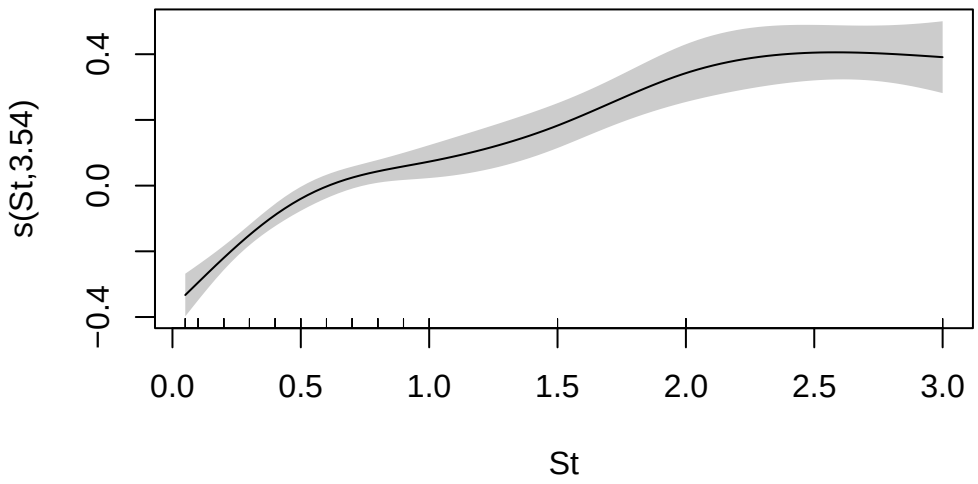
```
model_mean <- gam(
  log(R_moment_1) ~ Re + Fr + s(St, k = 5),
  data = train,
  method = "REML"
)
```

Inspect the fit

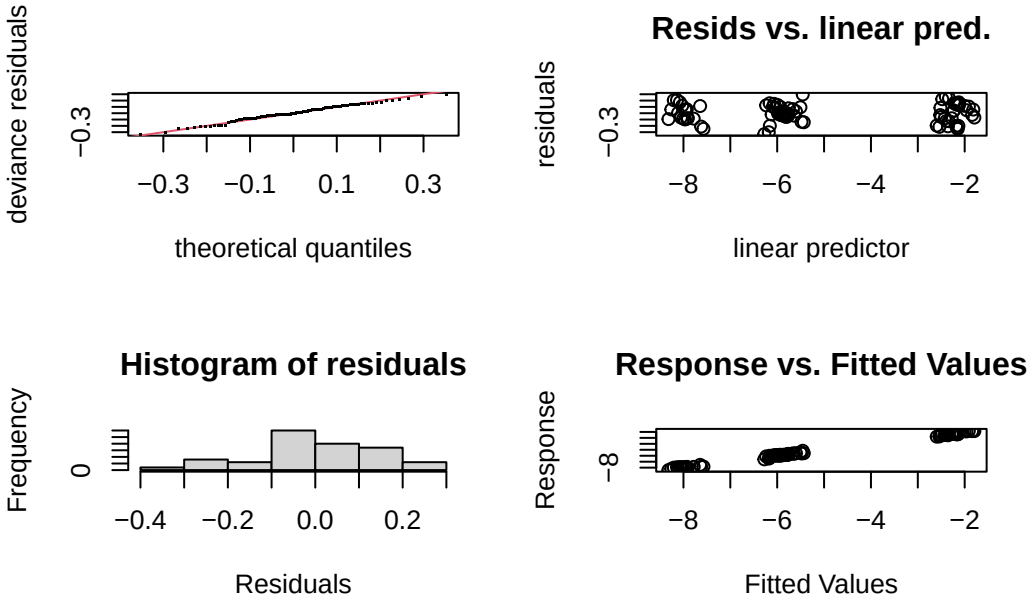
```
summary(model_mean)
```

```
Family: gaussian
Link function: identity

Formula:
log(R_moment_1) ~ Re + Fr + s(St, k = 5)

Parametric coefficients:
            Estimate Std. Error  t value Pr(>|t|)
(Intercept) -2.18389    0.03144  -69.466  < 2e-16 ***
Re224       -3.63949    0.03363 -108.213  < 2e-16 ***
Re398       -5.77972    0.04254 -135.864  < 2e-16 ***
Fr0.3       -0.11795    0.03871   -3.047  0.00313 **
FrInf       -0.02276    0.03450   -0.660  0.51138
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
        edf Ref.df     F p-value
s(St) 3.537  3.859 50.11  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.996   Deviance explained = 99.6%
-REML = -31.975  Scale est. = 0.019356  n = 89
```

```
plot(model_mean, shade = TRUE)
```




```
gam.check(model_mean)
```



```
Method: REML   Optimizer: outer newton
full convergence after 5 iterations.
Gradient range [-6.099059e-06,2.025272e-07]
(score -31.97488 & scale 0.01935551).
Hessian positive definite, eigenvalue range [0.5642027,41.53933].
Model rank =  9 / 9

Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

        k'  edf k-index p-value
s(St) 4.00 3.54    0.79    0.02 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```